

Hammad Ahmad

AI/ML Researcher | Trustworthy NLP · Retrieval-Augmented Generation · Knowledge Graphs

☎ +44 7555 856 155 | ✉ hammadahmad.ml@gmail.com | 📍 Bradford, UK

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#)

RESEARCH PROFILE

AI/ML researcher with an MSc (Merit) in Artificial Intelligence and a first-author peer-reviewed publication (Springer). My work centres on reliable, transparent language systems: retrieval-augmented generation grounded in knowledge graphs to reduce hallucination, evaluated against benchmarks I build and release, alongside predictive modelling on high-dimensional real-world health data. I am seeking a PhD to develop this into rigorous research on trustworthy, knowledge-grounded NLP.

EDUCATION

University of Bradford

Sep 2024 - Sep 2025

MSc, Artificial Intelligence (Merit)

Bradford, UK

- **Dissertation:** FinLaw-UK: A Graph-Augmented Retrieval Chatbot for Reliable and Transparent UK Financial Regulation.

COMSATS University Islamabad

Sep 2020 - Jul 2024

BS, Bioinformatics

Islamabad, Pakistan

- **Thesis:** AI-Assisted Analysis and Prediction of At-Risk Diabetic Individuals (final year research project graded A)

PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: [10.1007/978-3-031-66854-8_1](#)

RESEARCH EXPERIENCE

University of Bradford

Jan 2025 - Sep 2025

Research Assistant | Graph-Augmented LLM Engineering

Bradford, UK

- Designed and evaluated **FinLaw-UK**, a hybrid architecture combining retrieval-augmented generation, a Neo4j knowledge graph and a locally served Mistral 7B to reduce hallucination in UK financial-regulation question answering, grounded in the FCA Handbook, PRA Rulebook, FRC standards and statutory instruments.
- Constructed a 110-item evaluation benchmark (80 factual questions, 20 document tasks, 10 case scenarios) and assessed the pipeline with RAGAS alongside five custom lexical metrics, behind a strict graph-based verification gate that flagged and refused fabricated citations rather than answering.
- Quantified the cost of refusing: headline answer relevancy falls 23 points because RAGAS scores a refusal at zero by construction, but excluding the 30 refusals the mean exceeds the non-refusing baseline (0.658 against 0.641), which scored higher only by confabulating answers to unanswerable questions.
- Re-measured the submitted evaluation post-hoc and found two reported metrics were regex shape-checks, not correctness measures. Published the correction and the true graph-verified rate of 3 in 110.

COMSATS University Islamabad

Jul 2023 - Jul 2024

Research Intern, Data Science

Islamabad, Pakistan

- Benchmarked 11 classifiers for diabetes risk prediction on a 253,680-record CDC BRFSS dataset (14% positive class), comparing random over-sampling, SMOTE and ADASYN for class imbalance with resampling confined to the training folds; Random Forest performed best on ROC-AUC and sensitivity.
- Analysed 20+ demographic, lifestyle and clinical indicators, identifying age, general health, BMI, blood pressure and income as the strongest diabetes correlates, and deployed the model as a public risk-screening web app (React, Flask): **DiabetesSense**.
- Led a first-author comparative study of sleep-efficiency prediction across four regression models (best model Random Forest, R^2 0.86), published at [Springer](#).

SOFTWARE & ENGINEERING PROJECTS

Jobzyl | **Live Platform** | Data Pipelines & Semantic Search · Python, FastAPI, Postgres

Jun 2026 - Present

- Constructed and maintain a **2M+ document corpus** from 20 heterogeneous job-board and national employment-service APIs, with cross-source near-duplicate resolution and free-text compensation normalised onto a common annual basis.
- Implemented semantic vector ranking over CV-posting pairs with named skill-gap extraction, under an explicit abstention rule that withholds a score when a posting's extracted skills are too sparse to support one.

VoiceFlow | **GitHub** | Speech Transcription Pipeline · FastAPI, PyTorch, Whisper

Early 2026

- Built a reproducible speech-to-text pipeline transcribing **2,100+ production call recordings** on-device with Whisper large-v3, keeping identifiable customer audio inside the trust boundary and out of any third-party processor.
- Used the sequential long-form decoding path with per-file language identification and n -gram repetition blocking to suppress Whisper's degenerate looping on silence; run manifests make an interrupted export resumable.

INDUSTRY EXPERIENCE

Outlyst

Oct 2025 - Mar 2026

AI / Machine Learning Engineer

Leeds, UK (Remote)

- Developed backend dialogue-flow logic and workflow automation for an AI voice-agent system (Retell AI, FastAPI), integrating LLM-based dialogue with production infrastructure across 2,100+ calls.
- Reduced average end-to-end response latency 54%, from 2.4s to 1.1s, through async I/O profiling and connection pooling, a decisive factor in conversational systems where response delay degrades interaction quality.

Quality Control Assistant, Myton Food Group (Morrisons), Bradford, UK

Apr 2026 - Present

TECHNICAL SKILLS

Machine Learning & NLP : LLMs, Retrieval-Augmented Generation (RAG), Sentence Transformers, cross-encoder re-ranking, RAGAS evaluation, predictive modelling, ensemble methods, PyTorch, scikit-learn

Knowledge & Data : Neo4j knowledge graphs, vector embeddings, semantic search, PostgreSQL, pandas, NumPy

Programming & Tools : Python, SQL, TypeScript, FastAPI, Docker, Git, Linux, Ollama

REFERENCES

Dr Irfan Mehmood, Associate Professor in Business Analytics and Artificial Intelligence, University of Bradford

i.mehmood@bradford.ac.uk | +44 (0)1274 232646

Professor Tillal Eldabi, Professor of Business Analytics, University of Bradford | t.a.eldabi@bradford.ac.uk (*further referees available on request*)